



ADMAS UNIVERSITY

MEKANISA CAMPUS

DEPARTMENT OF COMPUTER

SCIENCE (CoSc)

Industrial Practice

By

- **Name - Bruk Tsegaye** **ID 0441/22**
- **Name of Industrial Practice hosting company - KNS Trading PLC**
- **Time duration if industrial Practice: June 20, 2025 - September 2, 2025**
- **Report Submission Date - September 5, 2025**

Executive Summary

This report presents a comprehensive summary of my two-month industrial practice program at **KNS Trading**, a company specializing in web development, hosting, IT solutions, and digital services. The primary objective of the training was to bridge the gap between academic knowledge and real-world industrial applications, equipping me with hands-on skills in modern web and application development.

During the practice, I was engaged in multiple domains of software and web technologies. I worked extensively with **Content Management Systems (Drupal and WordPress)** to design and manage dynamic websites. In addition, I explored **mobile application development** using **Flutter**, gaining valuable exposure to cross-platform design and deployment.

A significant portion of my work also involved **back-end and database-driven web development**. I learned and practiced **Laravel (PHP framework)** to build secure and scalable applications, while using **MySQL** for database management and optimization. Furthermore, I gained practical skills in **PHP scripting**, which strengthened my ability to handle server-side logic and data integration. To manage hosting environments and deployment processes, I was introduced to **cPanel**, where I carried out domain management, file handling, and server configuration tasks.

This diverse exposure not only enhanced my **technical expertise** but also improved my **problem-solving, teamwork, and communication skills**. Challenges, such as learning new frameworks (Laravel, Flutter) and adapting to industry-standard tools (cPanel, MySQL), were overcome through self-learning, mentorship, and collaboration.

In conclusion, my industrial practice at KNS Trading provided me with invaluable practical experience across **content management systems, web development frameworks, database administration, mobile application development, and hosting technologies**. The knowledge and skills gained have strengthened my readiness to contribute effectively to the IT industry and to pursue innovative solutions in future professional endeavors.

Declaration

I, the undersigned, hereby declare that this Industrial Practice Report is my original work and has not been submitted, in whole or in part, for the award of any degree or qualification at any other institution.

The report is based on my two-month industrial practice carried out at **KNS Trading**, where I was engaged in **web development and application development** using **Drupal, WordPress, Laravel, PHP, MySQL, Flutter, and cPanel**. The findings, reflections, and experiences presented in this report are genuine and drawn from the tasks I performed and the challenges I encountered during the training period.

Any data, materials, or information obtained from external sources have been properly acknowledged and referenced in accordance with academic integrity standards.

I take full responsibility for the content of this report.

Name of Student: Bruk Tsegaye

ID Number: 0441/22

Department: Computer Science

Signature: _____

Date: September 5, 2025

Approval of the Mentor

I, the undersigned, hereby certify that the above student has successfully completed the two-month industrial practice/apprenticeship program under my guidance and supervision. I have read and reviewed this report and approve it as a partial fulfillment of the requirements of the Industrial Practice Program.

Mentor's Name: Yinebeb Abebe

Position/Title: General Manager

Organization/Company: KNS Trading

Signature: _____

Date: September 5, 2025

Approval of the Supervisor

This is to certify that I have reviewed the apprenticeship report submitted by the student and I approve it as a valid and satisfactory presentation of the work undertaken during the industrial practice program.

Supervisor's Name: _____

Position/Title: _____

University/Department: _____

Signature: _____ Date: _____

Acknowledgment

First and foremost, I would like to express my sincere gratitude to **Mr. Yinebeb**, the owner of KNS Trading, for giving me the opportunity to take part in this apprenticeship program and for his constant support and encouragement throughout my training. His guidance and willingness to share valuable knowledge made this experience both insightful and rewarding.

My heartfelt appreciation also goes to **Mrs. Kelemuwa**, whose cooperation, patience, and mentorship greatly helped me in understanding the practical aspects of the work environment. Her assistance and constructive advice contributed significantly to my learning process.

I would also like to thank my fellow student colleagues who worked alongside me during this period. Their collaboration and teamwork created a positive and motivating atmosphere that made the training both enjoyable and productive.

Finally, I am truly grateful to everyone who contributed, directly or indirectly, to the success of my two-month industrial practice.

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1. Project Title

Federal Public Procurement Service (FPPS) Web Portal

1.1. Short Summary of the Project

The project focuses on the design and development of a digital platform for the **Federal Public Procurement Service (FPPS)**, which is responsible for ensuring transparency, efficiency, and accountability in public procurement activities. The system is intended to simplify procurement processes by enabling government agencies, suppliers, and other stakeholders to access information, submit bids, and monitor procurement activities through a centralized platform.

The project integrates the concept of **Content Management Systems (CMS)** using **Drupal** for dynamic website management, while also exploring **Flutter mobile application development** for extended accessibility. This approach ensures that procurement processes are not only transparent but also accessible to stakeholders across different devices.

1.2. Objective of the Project

- To design and implement a user-friendly web platform for managing public procurement services.
- To enhance **transparency** by publishing tenders, bids, and contract awards openly.
- To simplify and **automate procurement workflows**, reducing paperwork and delays.
- To provide stakeholders with real-time access to procurement information.
- To apply practical knowledge of **Drupal, and Flutter** in solving a real-world organizational problem.

1.3. Source of Information and Literature Review

The project was selected after analyzing the **challenges of public procurement systems in Ethiopia**, where manual processes and fragmented communication often lead to inefficiency, lack of transparency, and delays.

Primary Sources:

- Discussions with mentors at **KNS Trading** (Mr. Yinebeb and Mrs. Kelemwa), who provided insights into system requirements and practical implementation of CMS.
- Collaboration with fellow students during the apprenticeship, which helped refine the problem statement and technical approach.

Secondary Sources:

- Official documents and reports published by the **Federal Public Procurement Service (FPPS)**.
- Academic articles on **e-procurement systems**, transparency in governance, and ICT solutions for public administration. Online resources, tutorials, and developer documentation for **Drupal, and Flutter**, which guided the technical implementation process.

Related Literature:

Several studies highlight the importance of **e-governance** in improving public services. Research indicates that procurement systems powered by digital solutions reduce corruption risks, minimize administrative costs, and increase stakeholder trust. Furthermore, literature on **CMS platforms** emphasizes their role in providing scalable and customizable solutions for large organizations, while studies on **mobile app development** highlight the growing demand for cross-platform accessibility in governance systems

2. Background of the Hosting Company

2.1. Brief History

The hosting company where I carried out my industrial practice is a modern technology firm that specializes in **web development and IT solutions**. Established with the vision of providing innovative, affordable, and reliable digital services, the company has grown into a trusted partner for both governmental and private organizations. Over the years, it has built a reputation for delivering user-friendly, secure, and scalable systems that help organizations achieve digital transformation.

The company was founded by a team of passionate IT professionals who identified a gap in the Ethiopian market for professional, end-to-end web services. By continuously updating its technology stack and adopting international best practices, the company has positioned itself as one of the leading local firms in **web design, hosting, and IT consultancy**.

2.2. Main Products and Services

The company offers a wide range of products and services tailored to meet the needs of businesses and institutions in today's digital era. Its core services include:

- **Web Design & Hosting** – Creating responsive, user-friendly, and secure websites, along with reliable hosting solutions.
- **Database Solutions** – Designing and managing robust databases for organizations to securely store and process data.
- **Mobile and Web App Design** – Developing customized applications, including cross-platform apps, to enhance service delivery.
- **Network Installation & Maintenance** – Providing structured cabling, server configuration, and secure network setups.
- **IT Consultancy** – Offering expert advice on technology adoption, digital transformation, and IT infrastructure planning.
- **IT Support** – Delivering on-demand technical support for hardware, software, and systems management.
- **Digital Marketing** – Helping businesses reach wider audiences through SEO, social media management, and online campaigns.

2.3. Main Customers / End Users

The company's customers come from a diverse background, reflecting the broad applicability of its services:

- **Governmental Organizations** – Ministries, federal agencies, and public service institutions rely on the company for websites, secure databases, and digital systems.
- **Private Companies** – Businesses in sectors such as finance, education, manufacturing, and trade use the company's services to expand their digital presence and improve efficiency.

- **Small Enterprises & NGOs** – The company also provides affordable solutions for smaller organizations that want to grow their visibility and operations through IT tools.

2.4. Organizational Structure

The company follows a **flat but functional organizational structure**, ensuring smooth communication and efficiency.



Figure 1 Organizational Structure

2.5. Workflow

The company's workflow is designed to maximize flexibility and customer satisfaction. Its motto is: **“Anytime, Anywhere – Except on Sundays.”**

- **Client Engagement** – Customers can reach the company through online platforms, phone, or direct visits. Initial consultations are carried out to identify needs and propose solutions.
- **Project Assignment** – Based on requirements, the project manager assigns tasks to relevant teams (web design, app development, networking, etc.).
- **Development & Testing** – Teams work collaboratively, often using online collaboration tools to ensure progress is transparent and trackable.
- **Client Feedback Loop** – Prototypes or drafts are shared with clients for input and revisions.
- **Deployment & Support** – Once approved, the project is deployed, hosted, or delivered. IT support is available around the clock except on Sundays.

3. Overall Industrial Practice Experience and Specific Work

3.1. Why I Selected This Company

I chose to undertake my industrial practice at **KNS Trading**, a professional IT and web development company, because of its reputation for delivering high-quality and practical solutions in the field of **Web Design, Application Development, Database Management, and IT Consultancy**. The company is also known for its modern work culture and the opportunity it gives students to engage in real-world projects rather than just observation.

Another reason I selected this company is that it aligns with my academic interests in **web and mobile application development**. The chance to work with **Drupal, WordPress, and Flutter** provided me with hands-on experience in tools and technologies that are widely used in the industry but are often difficult to master without practical exposure.

3.2. Section of the Company I Worked In

During the apprenticeship, I was assigned to both the **Web Development Section** and the **Application Development Section**.

- In the **Web Development Section**, I was involved in managing and customizing websites built with **Drupal and WordPress**. This gave me the opportunity to understand CMS frameworks, how content is structured, and how user-friendly websites are built for both governmental and private clients.
- In the **Application Development Section**, I explored **Flutter app development**. Though challenging, it provided me with practical knowledge about cross-platform application development, UI/UX design, and coding practices for mobile applications.

This dual placement helped me gain a balanced skill set in both web and mobile development

3.3. Work Tasks Executed

- **Website Translation into Amharic** – One of my main tasks was translating an existing Drupal-based website into Amharic. This was particularly important because many end-users prefer localized content for better accessibility. This task taught me how to manage multilingual features in a CMS and configure language modules.
- **App Development** – I worked on developing a simple Flutter application. My focus was on learning the fundamentals of Flutter, building UI components with widgets, and ensuring the app was functional and user-friendly. I practiced by creating small apps, such as a calculator, and explored ways to scale them into larger projects.

3.4. Challenges and Problems Faced

- **New to Drupal CMS** – Since I had no prior practical experience with Drupal, understanding its architecture, modules, and content structure was initially overwhelming.
- **New to Flutter** – Developing mobile apps in Flutter was also a new experience. Understanding widgets, debugging, and state management was challenging.
- **Time Constraints** – Since the industrial practice was limited to two months, I had to balance learning and execution within a short period.
- **Language Barriers in Translation** – Translating technical content into Amharic required accuracy and attention to detail to maintain meaning without losing professionalism.

3.5. Measures Taken to Overcome Challenges

To overcome these challenges, I adopted several measures:

- **Self-Learning** – I spent extra time outside of office hours learning through online tutorials, developer documentation, and video lessons.
- **Asking Questions** – I frequently consulted my mentors (Mr. Yinebeb and Mrs. Kelemuwa) whenever I faced technical roadblocks. Their explanations helped me understand faster.

- **Trial and Error** – For Flutter, I practiced by building small projects first before attempting larger ones. This incremental approach helped me reduce mistakes.
- **Team Collaboration** – I worked with fellow students to share knowledge and solve problems together.

These measures not only helped me complete my tasks successfully but also improved my problem-solving and research skills.

3.6. Recommendations Made Regarding Identified Problems

Based on my experience, I made the following recommendations:

- **Provide Beginner-Friendly Documentation** – The company should prepare simple guides and checklists for students or new employees to quickly understand Drupal and Flutter.
- **Introduce a Mentorship Schedule** – Assigning short, structured mentoring sessions each week would help apprentices stay on track and avoid unnecessary delays.
- **Create a Practice Environment** – Setting up a sandbox environment where students can freely practice CMS and app development without affecting live projects would make learning smoother.
- **Promote Knowledge Sharing** – Weekly knowledge-sharing sessions among team members and apprentices could enhance collaboration and problem-solving.
- **Language Localization Tools** – For translation projects, the company could integrate advanced multilingual plugins or AI-powered tools to ensure accuracy and save time.

Conclusion

Overall, my industrial practice was both **challenging and rewarding**. I not only improved my technical knowledge in **Drupal, WordPress, and Flutter**, but also developed soft skills such as problem-solving, teamwork, and adaptability. The experience has significantly contributed to my professional growth, giving me the confidence to tackle real-world projects and preparing me for future career opportunities in web and mobile development.

4. Overall Benefits Gained from Industrial Practice

My two-month industrial practice at **KNS Trading** has been a transformative learning experience that bridged the gap between academic knowledge and real-world industry application. Through active participation in web development, app development, and translation tasks, I gained both technical and non-technical skills that will support my career as a computer science professional. The overall benefits of this industrial practice can be summarized in the following dimensions:

1. Enhancement of Hands-On Technical Experience

I was able to transform theoretical knowledge into practical application by working directly with technologies such as **Drupal, WordPress, Laravel, PHP, MySQL, Flutter, and cPanel**. This exposure allowed me to gain confidence in building, managing, and deploying real-world web and mobile applications.

2. Upgrading of Theoretical Knowledge

Concepts learned in the classroom—such as database design, object-oriented programming, and system deployment—were deepened through real-world practice. This connection between theory and practice enhanced my understanding of how software development works in an industrial setting.

3. Strengthening of Analytical and Technical Problem-Solving Skills

I faced real challenges while working with frameworks like **Laravel** and tools like **cPanel**, which required me to analyze problems critically and find practical solutions. This experience sharpened my ability to troubleshoot errors, optimize systems, and adapt to new technologies.

4. Team Collaboration and Communication Skills

Working with professionals and fellow trainees improved my teamwork abilities. I learned how to communicate technical ideas clearly, share responsibilities effectively, and coordinate efforts to complete projects within deadlines.

5. Development of Leadership Qualities

I gradually took initiative in small project components, guiding peers, and suggesting solutions. This experience enhanced my ability to take responsibility and demonstrate leadership in a professional environment.

6. Understanding Work Ethics and Professional Conduct

The industrial practice taught me the importance of punctuality, discipline, accountability, and respecting organizational culture. These values helped me grow both professionally and personally.

7. Entrepreneurial and Innovative Mindset

Exposure to client-based projects inspired me to think creatively about problem-solving and potential business opportunities in the IT field.

8. Improved Interpersonal Communication Skills

By engaging with mentors, coworkers, and clients, I refined my communication skills—both verbal and written—which will be vital for my future career.

Comparative Gains from Different Technologies

1. Content Management Systems (Drupal & WordPress)

- Learned how to **design, customize, and manage dynamic websites** efficiently.
- Gained skills in **theme customization, plugin integration, and multilingual support** (e.g., translating content into Amharic).
- Improved understanding of **non-technical client requirements**, since CMS platforms are often tailored to ease of use.
- Built confidence in **rapid prototyping and content publishing** without deep coding.

2. Laravel, PHP & MySQL

- Strengthened my **backend development skills** using PHP and Laravel framework.
- Learned to design and implement **relational databases** with MySQL, improving my data management and query optimization skills.
- Understood **MVC (Model-View-Controller)** architecture and how it improves scalability and maintainability.
- Gained practical experience in **server-side programming**, authentication, and database-driven applications.
- Enhanced ability to work with **cPanel** for deployment, hosting, and server management.

3. Flutter (Cross-Platform Mobile Development)

- Learned to build **mobile applications for both Android and iOS** from a single codebase.
- Gained experience in **UI/UX design** using Flutter's widget-based approach.
- Understood how to integrate features such as **navigation, state management, and data handling** in mobile apps.
- Developed adaptability in learning **new frameworks** that differ from traditional web development.
- Strengthened **problem-solving skills** while dealing with framework-specific challenges and debugging mobile environments.

5. General Conclusion and Recommendations

5.1. General Conclusion

The two-month industrial practice at **KNS Trading** has been an invaluable experience that bridged the gap between academic learning and real-world application. Throughout the training, I had the opportunity to engage with diverse technologies, including **Drupal, WordPress, Laravel, PHP, MySQL, Flutter, and cPanel**, which gave me both breadth and depth in the field of web and mobile application development.

The practice strengthened my ability to design, develop, and manage projects in a professional environment, while also improving my adaptability to new tools and frameworks. I was able to transform theoretical concepts into practical solutions, gain confidence in handling technical challenges, and enhance my analytical and problem-solving skills. Moreover, the experience helped me understand the importance of teamwork, professional ethics, and effective communication in achieving organizational goals.

One of the most significant outcomes of this practice is the holistic growth it offered. Beyond technical expertise, I gained exposure to leadership, innovation, and entrepreneurship—qualities that will support my career advancement in the IT industry. The challenges I encountered, such as adapting to new frameworks and

troubleshooting deployment issues, served as valuable learning experiences that prepared me for the dynamic nature of the technology sector.

In conclusion, my industrial practice at KNS Trading was not only a stepping stone toward professional readiness but also a foundation for lifelong learning and growth. I am confident that the knowledge, skills, and values I acquired during this period will guide me in making meaningful contributions to both the academic and industrial worlds in the future.

5.2. Recommendations for the Company

Based on my observations and personal experience, I propose the following recommendations that can help the company improve its operations and training approach:

1. **Structured Training for Apprentices** – Develop short training modules or orientation sessions for new students/apprentices, especially on tools like Drupal and Flutter, to reduce the initial learning curve.
2. **Documentation and Knowledge Sharing** – Prepare simplified internal manuals and technical guidelines for frequently used systems. Weekly knowledge-sharing sessions among staff and apprentices would also strengthen collaboration.
3. **Innovation in Services** – Explore opportunities to expand into **AI-assisted development, cloud hosting, and cybersecurity services**, which are increasingly important for both governmental and private clients.
4. **Investment in Localization** – Since language accessibility is crucial in Ethiopia, the company should invest in **multilingual solutions** and **AI-powered translation tools** to improve user experience.
5. **Client Engagement and Feedback** – Establish a formal mechanism for gathering feedback from clients after project delivery. This will help improve quality and build stronger relationships.
6. **Encouraging Entrepreneurship** – Provide mentorship sessions where senior developers and managers share their journey and entrepreneurial insights with students. This will inspire apprentices to pursue innovation

7. Workplace Development – Continue fostering a positive workplace culture with attention to **industrial psychology** (motivation, teamwork, recognition) to increase productivity and employee satisfaction.

In conclusion, the company has already built a strong foundation in **Web Development, IT Consultancy, and Digital Solutions**. By adopting these recommendations, it can further strengthen its position as a leading technology provider while also becoming a better training ground for future IT professionals.

6. References

- Federal Public Procurement Service (FPPS) official website and documents.
- KNS Trading Company internal documents, project guidelines, and training materials.
- Drupal Documentation: <https://www.drupal.org/documentation>
- WordPress Developer Handbook: <https://developer.wordpress.org>
- Flutter Official Documentation: <https://docs.flutter.dev>
- Academic articles on e-Government, e-Procurement, and CMS usage in developing countries.
- Mentorship discussions with Mr. Yinebeb (Owner, KNS Trading) and Mrs. Kelemuwa (Project Coordinator, KNS Trading)